

Lec 7-1, Oct 20, 2025.

- Fundamental theorems of calculus (FTC, 5.4)
- Areas between curves.
- Substitution methods (5.5, 5.6, also known as change of variable)

• Fundamental Theorems of Calculus. [FTC]

Let $v(t)$ be the velocity function of an object, say $v(t) \geq 0, \forall t$.

Q1. What is $\int_1^5 v(t) dt$?

A1. Total displacement travelled from $t=1$ to $t=5$.

Q2. What is $F(x)$ if $F(x) \triangleq \int_1^x v(t) dt$?

A2. Total displacement travelled from $t=1$ to $t=x$.

Q3. What is $F'(x)$?

A3. Marginal movement at $t=x$

↳ velocity at $x = v(x)$.

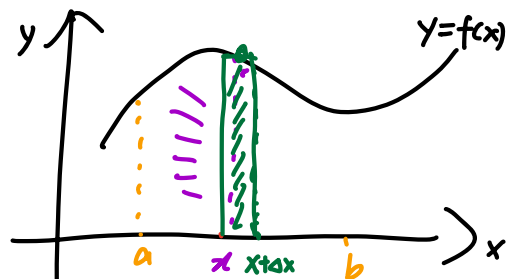
• Look at a more general case.

— Suppose f is continuous on $[a, b]$.

— Let $F: [a, b] \rightarrow \mathbb{R}$, $\underline{F(x)} \triangleq \int_a^x f(t) dt$

Q: What is $F'(x)$ for $x \in (a, b)$?

A: $f(x)$.



$$\frac{F(x+\Delta x) - F(x)}{\Delta x} \approx f(x) \cdot \Delta x$$

• FTC 1

THEOREM 4—The Fundamental Theorem of Calculus, Part 1 If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$:

$$F'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x). \quad (2)$$

$$F'_+(a) = f(a), \quad F'_-(b) = f(b).$$

[Also works for $x=a$ and b with one-sided derivatives.]

Pf: we will show $F'(x) = f(x)$ for $x \in (a, b)$, $F'_+(a) = f(a)$ and $F'_-(b) = f(b)$.

Then differentiability implies continuity (i.e. F is cont on $[a, b]$).

$$\begin{aligned} \textcircled{1} \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} &= \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(t) dt - \int_a^x f(t) dt}{h} \\ &= \lim_{h \rightarrow 0} \frac{\int_x^{x+h} f(t) dt}{h} \end{aligned}$$

[This is the average value of f between x and $x+h$, even if $h < 0$]

$\textcircled{2}$ By MVT for integrals, $\exists c_h$ between x and $x+h$, s.t.

$$\frac{\int_x^{x+h} f(t) dt}{h} = f(c_h).$$



$\textcircled{3}$ Note that $c_h \rightarrow x$ as $h \rightarrow 0$

$\therefore$ By $\textcircled{1}$ and $\textcircled{2}$, we have

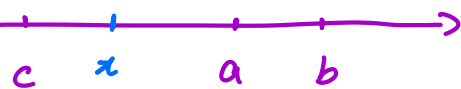
$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} f(c_h) = f\left(\lim_{h \rightarrow 0} c_h\right) = f(x).$$

$\therefore$ Similarly, $F'_+(a) = f(a)$ and $F'_-(b) = f(b)$.

□

Rmk. Suppose f is cont' on $[c, b]$,

where $c < a$. Define $G(x) \triangleq \int_a^x f(t) dt$, $x \in [c, b]$.



Q: Is $G'(x) = f(x)$ still true for $x \in (c, a)$?

A: Yes. $G(x) = \int_c^x f(t) dt + \int_a^c f(t) dt$ $(F(x) \triangleq \int_a^x f(t) dt)$

$$= F(x) + \int_a^c f(t) dt$$
$$\stackrel{\circ}{=} \underline{F(x)} + L.$$

$$\therefore G'(x) = F'(x) = f(x), \quad \forall x \in [c, b]. \quad \square$$

Rmks about FTC 1

① $\int_a^x f(\underline{t}) dt = \int_a^x f(s) ds$: the inner variable is a dummy variable,
so the choice is not important. Do NOT write $\int_a^x f(x) dx$.

② FTC 1 implies that any continuous function on I must have an antiderivative on I :

$F(x) \triangleq \int_a^x f(t) dt$ is an antiderivative of f .

EXAMPLE 2Use the Fundamental Theorem to find dy/dx if

(a) $y = \int_a^x (t^3 + 1) dt$

(b) $y = \int_x^5 3t \sin t dt$

(c) $y = \int_1^{x^2} \cos t dt$

(d) $y = \int_{1+3x^2}^4 \frac{1}{2+t} dt$

Sol: (a) Let $F(x) = \int_a^x (t^3 + 1) dt$. Then $y = F(x)$.

$$\therefore \frac{dy}{dx} = F'(x) \stackrel{\text{FTC1}}{=} f(x) = x^3 + 1.$$

(b) Let $F(x) = \int_5^x 3t \sin t dt$. Then $y = F(x)$.

$$\therefore \frac{dy}{dx} = F'(x) \stackrel{\text{FTC1}}{=} f(x) = 3x \sin x.$$

Technique: If $F(u) = \int_a^u f(t) dt$, then

$$\int_a^{g(x)} f(t) dt = F(g(x)),$$

and

$$\begin{aligned} \frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] &= \frac{d}{dx} [F(g(x))] \\ &= F'(g(x)) \cdot g'(x) \\ &\stackrel{\text{FTC1}}{=} f(g(x)) \cdot g'(x). \end{aligned}$$

(c) Let $F(u) = \int_1^u \underbrace{\cos t}_{f(t)} dt$ and $g(x) = x^2$.

Then $y = F(g(x))$

$$y = \int_1^{x^2} \cos t dt$$

$$\therefore \frac{dy}{dx} = [F(g(x))]' = F'(g(x)) \cdot g'(x)$$

$$\stackrel{\text{FTC1}}{=} f(g(x)) g'(x) = \cos(x^2) \cdot 2x.$$

$$= \cos(x^2) \cdot 2x.$$

(d). Let $F(u) = \int_4^u \underbrace{\frac{1}{2+t}}_{f(t)} dt$ and $g(x) = 1+3x^2$.

Then $y = -\int_4^{1+3x^2} \frac{1}{2+t} dt = -F(g(x)).$

$$\therefore \frac{dy}{dx} = -[F(g(x))]'$$

$$= -F'(g(x)) g'(x)$$

$$\stackrel{\text{FTC1}}{=} -f(g(x)) g'(x) = -\frac{1}{2+g(x)} \cdot 6x$$

$$= -\frac{6x}{3+3x^2} = -\frac{2x}{1+x^2}.$$

□.

• FTC 2.

- Let $v(t)$ be the velocity function.
- $\int_a^b v(t) dt = \text{total displacement travelled from } t=a \text{ to } t=b.$
- If $s(t)$ is the position at time t , then by physical intuition,

$$\int_a^b v(t) dt = s(b) - s(a).$$

- More generally,

THEOREM 4 (Continued)—The Fundamental Theorem of Calculus, Part 2 [FTC2]
If f is continuous over $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Pf: Step 1. Let $G(x) \triangleq \int_a^x f(t) dt$, $x \in [a, b]$.

By FTC1, $G'(x) = f(x)$. $\forall x \in (a, b)$, $G'_+(a) = f(a)$, $G'_-(b) = f(b)$,
which means that G is an antiderivative of f .

Step 2. Compute:

$$G(a) = 0,$$

$$G(b) = \int_a^b f(t) dt.$$

Step 3. Let F be any antiderivative of f .

Then we have

$$F(x) = G(x) + C, \quad \forall x \in [a, b].$$

Step 4. Compute:

$$F(b) - F(a) = [G(b) + C] - [G(a) + C]$$

$$= G(b) - G(a)$$

$$= \int_a^b f(t) dt - 0 = \int_a^b f(t) dt, \quad \square$$

* By FTC 2, in order to calculate $\int_a^b f(x) dx$,

it suffices to find an antiderivative F of f .

Rmk. ① FTC 1 & 2 imply that differentiation and integration are "inverse operations".

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

[FTC 1]

$$\text{and } \int_a^x f(t) dt = F(x) + C$$

[FTC 2]

$\Rightarrow$ unifying differentiation and integration [Newton-Leibniz Formula].

$$d \left[\underbrace{\int_a^x f(t) dt}_{\text{integral}} \right] = d F(x) = f(x) \underbrace{dx}_{\text{differential}}$$

② Applying differentiation and integration one after another

gives the original function back, up to a constant shift.

$$\int_a^x \left[\frac{d}{dt} F(t) \right] dt = \int_a^x F'(t) dt = F(x) + C.$$

↑
FTC2

□.

Notation : We write $F(x)|_{x=a}^b$ or $F(x)|_a^b$ to mean $F(b) - F(a)$.

e.g. ① $\int_0^\pi \cos x \, dx = \sin x \Big|_0^\pi = \sin \pi - \sin 0 = 0$.

② $\int_0^1 x^2 \, dx = \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n} \right) = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3} - 0 = \frac{1}{3}$.

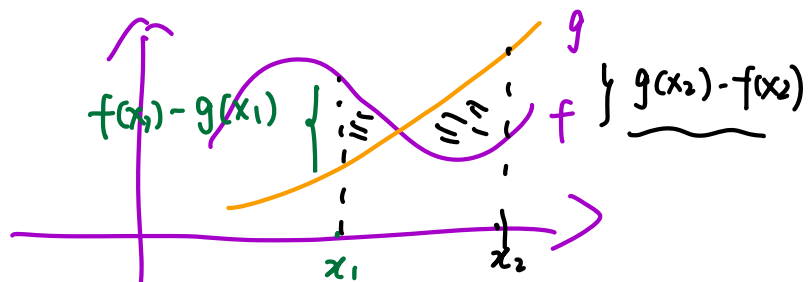
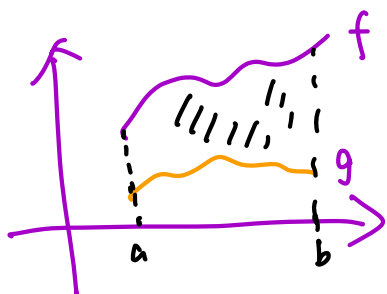
③ $\int_1^9 \frac{x-1}{\sqrt{x}} \, dx = \int_1^9 (x^{\frac{1}{2}} - x^{-\frac{1}{2}}) \, dx$
 $= \frac{2}{3} x^{\frac{3}{2}} \Big|_1^9 - (2x^{\frac{1}{2}}) \Big|_1^9$
 $= \frac{2}{3} x^{\frac{3}{2}} \Big|_1^9 - (2x^{\frac{1}{2}}) \Big|_1^9$
 $= \frac{2}{3} (9^{\frac{3}{2}} - 1) - (2 \cdot 3 - 2 \cdot 1) = \dots \quad \square$

Areas between curves

Definition

Let f and g be functions that are integrable on $[a, b]$. Then the **area** A between the graph of $y = f(x)$ and the graph of $y = g(x)$, from $x = a$ to $x = b$, is defined by

$$A := \int_a^b |f(x) - g(x)| \, dx.$$



- For area between $y = f(x)$ and the x -axis, take $g(x) \equiv 0$, and the area becomes

$$A = \int_a^b |f(x)| dx.$$

- If f is further nonnegative in the case above, then

$$A = \int_a^b f(x) dx,$$

which is the same as our previous intuition.

Rmk. The area is always nonnegative and the definite integral could be negative. Just use the **area** as one special perspective to understand the definite integral; but they are not fully equivalent.

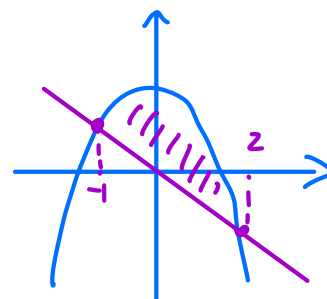
5.6.4.
EXAMPLE 4

Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

Sol¹: ① Let $2 - x^2 = -x$

Then $x^2 - x - 2 = (x - 2)(x + 1) = 0$

$\Rightarrow x_1 = 2, \text{ or } x_2 = -1$



② Area = $\int_{-1}^2 | (2 - x^2) - (-x) | dx$

= $\int_{-1}^2 (2 - x^2 + x) dx$

= $\left[-\frac{1}{3}x^3 + \frac{1}{2}x^2 + 2x \right]_{-1}^2$

= $\left[-\frac{8}{3} + \frac{1}{2} \times 4 + 2 \times 2 \right] - \left[+\frac{1}{3} + \frac{1}{2} - 2 \right]$

= $6 - 3 - \frac{1}{2} + 2$

= $\frac{9}{2}$.

EXAMPLE 5 ^{5.6.5.}

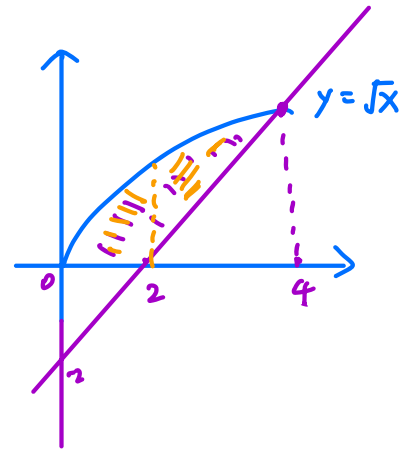
Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$.

Sol' ① Let $\sqrt{x} = x - 2$

$$\therefore x = (x-2)^2$$

$$\therefore x^2 - 5x + 4 = 0$$

$$\therefore x = 4 \text{ or } x = 1 \text{ (remove!)}$$



$$\textcircled{2} \text{ Area} = \int_0^2 \sqrt{x} \, dx + \int_2^4 (\sqrt{x} - (x-2)) \, dx$$

$$= \left. \frac{2}{3} x^{\frac{3}{2}} \right|_0^2 + \left. \frac{2}{3} x^{\frac{3}{2}} \right|_2^4 - \left. \frac{1}{2} x^2 \right|_2^4 + 2x \Big|_2^4$$

$$= \frac{2}{3} (2^{\frac{3}{2}} - 0) + \frac{2}{3} (4^{\frac{3}{2}} - 2^{\frac{3}{2}}) - \frac{1}{2} (4^2 - 2^2) + 2(4-2)$$

$$= \frac{2}{3} (4^{\frac{3}{2}} - 0) - \frac{1}{2} (12) + 2 \cdot 2$$

$$= \frac{2}{3} \times 8 - 6 + 4$$

$$= \frac{10}{3} .$$

□

Example

Find the area A between the graph of $y = f(x)$ and the graph of $y = g(x)$, from $x = a$ to $x = b$.

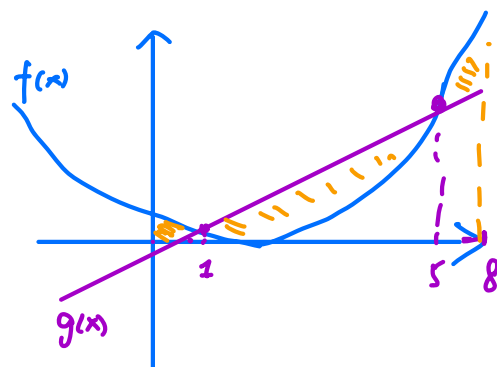
(c) $f(x) := (x-2)^2$ and $g(x) := 2x-1$; $a = 0$ and $b = 8$;

Sol': Let $(x-2)^2 = 2x-1$

Then $x^2 - 4x + 4 = 2x - 1$

$$x^2 - 6x + 5 = (x-1)(x-5) = 0$$

$\therefore x_1 = 1$, or $x_2 = 5$.



$$\therefore A = \int_0^8 |f(x) - g(x)| dx$$

$$= \int_0^1 (f(x) - g(x)) dx + \int_1^5 (g(x) - f(x)) dx + \int_5^8 (f(x) - g(x)) dx$$

$$= \int_0^1 ((x-2)^2 - (2x-1)) dx + \int_1^5 (2x-1 - (x-2)^2) dx + \int_5^8 ((x-2)^2 - (2x-1)) dx$$

$$= \int_0^1 (x^2 - 6x + 5) dx + \int_1^5 (-1)(x^2 - 6x + 5) dx + \int_5^8 (x^2 - 6x + 5) dx$$

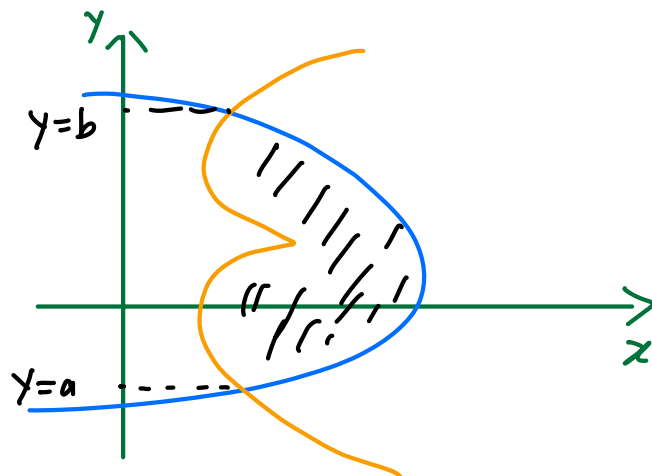
$$= 40.$$

□

Rmk. Area A between curves $x = f(y)$ and $x = g(y)$

from $y = a$ to $y = b$ can be defined similarly:

$$A = \int_a^b |f(y) - g(y)| dy$$



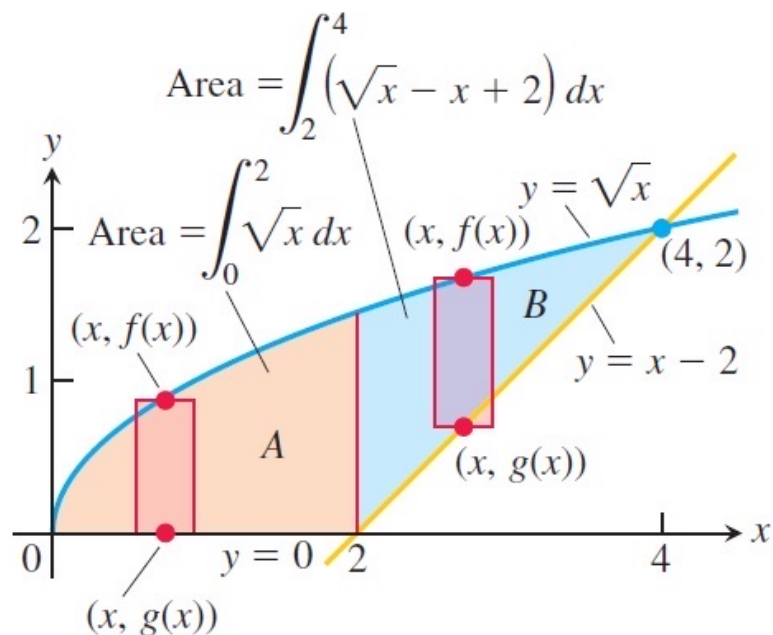


FIGURE 5.28 When the formula for a bounding curve changes, the area integral changes to become the sum of integrals to match, one integral for each of the shaded regions shown here for Example 6.

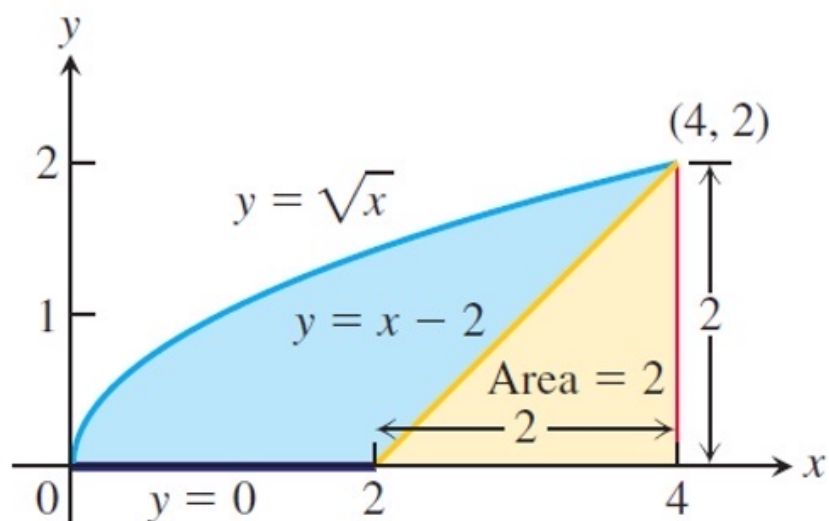
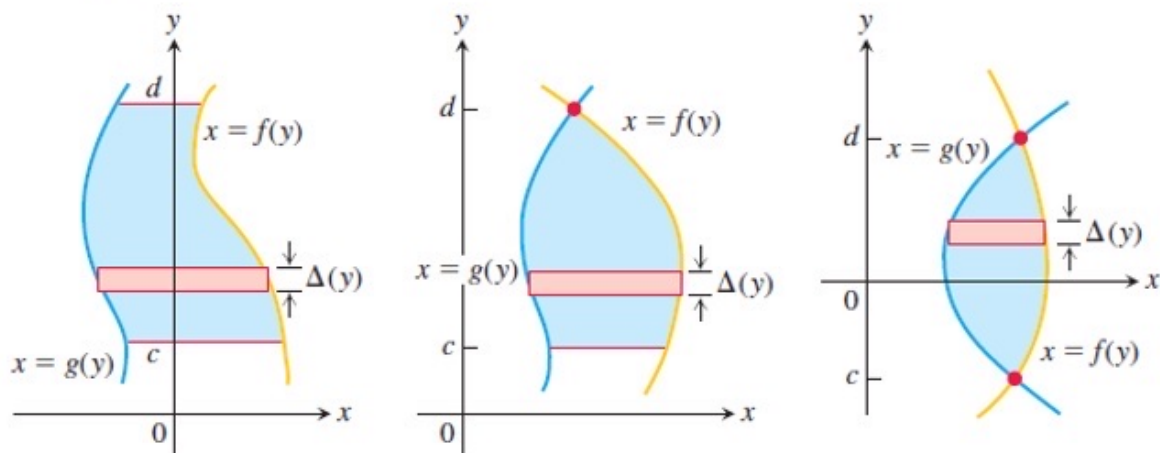


FIGURE 5.30 The area of the blue region is the area under the parabola $y = \sqrt{x}$ minus the area of the triangle.

Integration with Respect to y

If a region's bounding curves are described by functions of y , the approximating rectangles are horizontal instead of vertical and the basic formula has y in place of x .

For regions like these:



use the formula

$$A = \int_c^d [f(y) - g(y)] dy.$$

In this equation f always denotes the right-hand curve and g the left-hand curve, so $f(y) - g(y)$ is nonnegative.

FTC 2 states that in order to compute $\int_a^b f(x) dx$, it suffices to find a computable antiderivative F of f .

In the second half of the course, we are going to introduce a few techniques of finding F .

The first is the substitution method.

• Substitution method (change of variable)

e.g. What is an antiderivative of $f(x) = \frac{1}{\sqrt{x}} \cos \sqrt{x}$?

Sol': $\therefore$ To solve F , s.t. $F' = f$.

$$\begin{aligned} \cdot \underbrace{F(x)}_{\Delta} &= \int f(x) dx + C \\ &= \int \frac{1}{\sqrt{x}} \cos \sqrt{x} dx + C \\ &= \int \cos \sqrt{x} \cdot \underbrace{\frac{1}{\sqrt{x}} dx}_{d(2x^{\frac{1}{2}})} + C \\ &= \int \cos \sqrt{x} d(2x^{\frac{1}{2}}) + C \\ &\stackrel{u=2\sqrt{x}}{=} \int \cos\left(\frac{u}{2}\right) du \Big|_{u=2\sqrt{x}} + C \\ &= 2 \sin \frac{u}{2} \Big|_{u=2\sqrt{x}} + C \\ &= 2 \sin(\sqrt{x}) + C \quad \square \end{aligned}$$

$$(2\sqrt{x})' = \frac{1}{\sqrt{x}}$$

$$\Rightarrow d(2\sqrt{x}) = \frac{1}{\sqrt{x}} dx$$

check:

$$\begin{aligned} &(2 \sin(\sqrt{x}))' \\ &= 2 \cos(\sqrt{x}) \cdot \frac{1}{2} x^{-\frac{1}{2}} \\ &= \cos(\sqrt{x}) \cdot \frac{1}{\sqrt{x}} \quad \checkmark \end{aligned}$$

5.5.6

THEOREM 8—The Substitution Rule If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

It means LHS = RHS
when both are expressed
in terms of x .

$$\int \underbrace{f(g(x))}_u \underbrace{g'(x) dx}_{du} = \int f(u) du.$$

$$= \int f(u) du \Big|_{u=g(x)} = F(g(x)) + C.$$

One may write $d(g(x))$ instead of du for $u = g(x)$.

$$\begin{aligned} \int f(g(x)) g'(x) dx &= \int f(g(x)) d(g(x)) \\ &\stackrel{u=g(x)}{=} \int f(u) du \Big|_{u=g(x)} \\ &= F(u) \Big|_{u=g(x)} + C. \end{aligned}$$

Before proving it, let us demonstrate with one example.

e.g. 1. $\int \sin^3 x dx = \int \sin^2 x \cdot \sin x dx$

$$\begin{aligned} &= \int \sin^2 x d(-\cos x) \\ &= -\int \sin^2 x d\cos x \\ &= \int (\cos^2 x - 1) d\cos x \end{aligned}$$

$$\stackrel{u=\cos x}{=} \int (u^2 - 1) du \Big|_{u=\cos x}$$

$$= \frac{1}{3} u^3 - u + C \Big|_{u=\cos x} = \frac{1}{3} \cos^3 x - \cos x + C. \quad \square$$

Check:

$$\begin{aligned} &\left[\frac{1}{3} \cos^3 x - \cos x + C \right]' \\ &= \frac{1}{3} \cdot 3 \cos^2 x (-\sin x) + \sin x + 0 \\ &= -\cos^2 x \sin x + \sin x \\ &= \sin x (1 - \cos^2 x) \\ &= \sin^3 x. \end{aligned}$$

Always check by
doing differentiation.